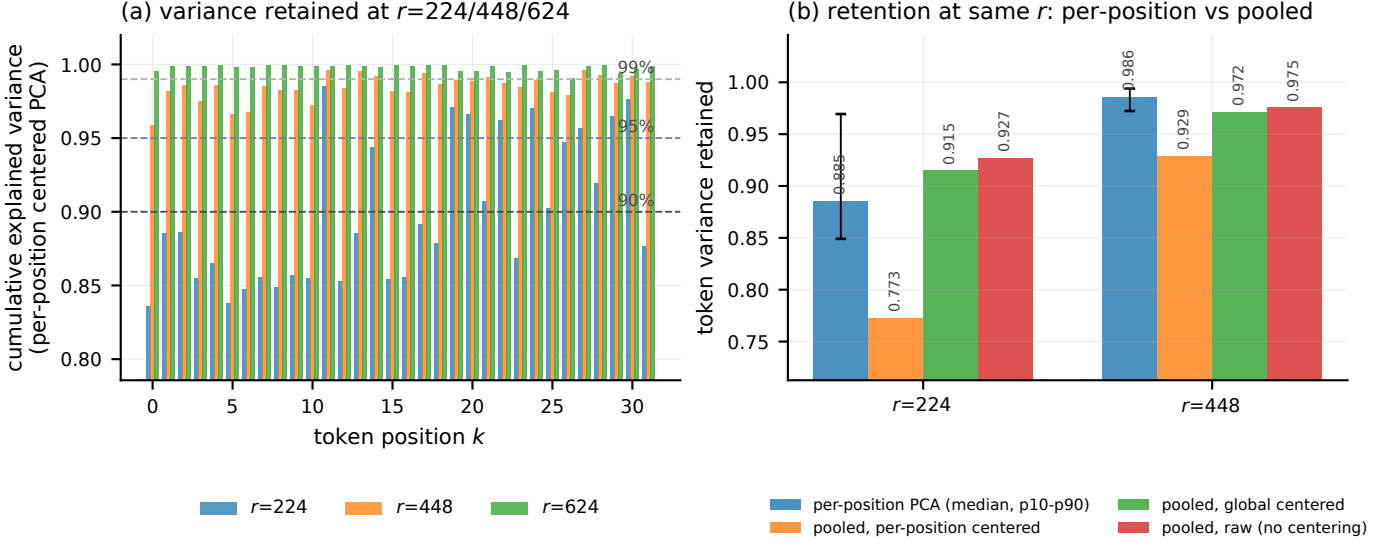
(a) variance retained at r=224/448/624
cumulative explained variance (per-position centered PCA)
1.00
0.95
0.90
0.85
0.80
99%
95%
90%
0
5
10
15
20
25
30
token position k
r=224
r=448
r=624
(b) retention at same r: per-position vs pooled
token variance retained
1.00
0.95
0.90
0.85
0.80
0.75
0.885
0.773
0.915
0.927
0.986
0.929
0.972
0.975
r=224
r=448
per-position PCA (median, p10-p90)
pooled, per-position centered
pooled, global centered
pooled, raw (no centering)